

Java Exception Programs

Codes and Respective Terminal Outputs

1. ArrayStoreException

Code

```
package exceptions;
public class ASEException {
    public static void main(String[] args) {
        Object[] arr = new String[3];
        try {
            arr[0] = "Java";
            arr[1] = "Python";
            arr[2] = 10; // it is an integer
        }
        catch (ArrayStoreException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Terminal Output

```
$ java Program
java.lang.Integer
```

2. ClassCastException

Code

```
package exceptions;
public class CcException {
    public static void main(String[] args) {
        Object obj = "aa";
        try {
            // aa is not a integer
            Integer num = (Integer) obj;
            System.out.println(num);
        }
        catch(ClassCastException e) {
            System.out.println("invalid type casting");
        }
    }
}
```

Terminal Output

```
$ java Program
invalid type casting
```

3. ArrayIndexOutOfBoundsException

Code

```
package exceptions;
class GFG {
    public static void main(String[] args) {
        try {
            int array[] = new int[] { 4, 1, 2, 6, 7 };
            System.out.println("The element at index 4 is "
                               + array[4]);
            System.out.println("The element at index 6 is "
                               + array[6]);
        }
        catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Terminal Output

```
$ java Program
The element at index 4 is 7
java.lang.ArrayIndexOutOfBoundsException: Index 6 out of bounds for length 5
```

4. IllegalArgumentException

Code

```
package exceptions;
public class IAEException {
    static void setAge(int age) {
        if (age<0) {
            throw new IllegalArgumentException("Age cannot be negative");
        }
        System.out.println("Age:"+age);
    }
    public static void main(String[] args) {
        try {
            setAge(20);
            setAge(-5);
        }
        catch (IllegalArgumentException e) {
            System.out.println("Error: " + e.getMessage());
        }
        System.out.println("Program continues");
    }
}
```

Terminal Output

```
$ java Program
Age:20
Error: Age cannot be negative
Program continues
```

5. IllegalStateException

Code

```
package exceptions;
class Printer {
    boolean on = false;
    void print() {
        if (!on) {
            throw new IllegalStateException("Printer is turned off");
        }
        System.out.println("Printing...");
    }
}
public class ISEException {
    public static void main(String[] args) {
        Printer p = new Printer();
        try {
            p.print();
        }
        catch (IllegalStateException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Terminal Output

```
$ java Program
Printer is turned off
```

6. NegativeArraySizeException

Code

```
package exceptions;
public class NASException {
    public static void main(String[] args) {
        try {
            int size = -5;
            int[] numbers = new int[size];
            System.out.println("Array created");
        }
        catch (NegativeArraySizeException e) {
            System.out.println("Array size cannot be negative");
        }
    }
}
```

Terminal Output

```
$ java Program
Array size cannot be negative
```

7. NumberFormatException

Code

```
package exceptions;
class NumberFormat_Demo {
    public static void main(String args[]) {
        try {
            int num = Integer.parseInt("akki 100");
            System.out.println(num);
        } catch(NumberFormatException e) {
            System.out.println("Number format exception");
        }
    }
}
```

Terminal Output

```
$ java Program
Number format exception
```

8. NullPointerException

Code

```
package exceptions;
class Student {
    String name;
}
public class NPE {
    public static void main(String[] args) {
        Student s = null;
        System.out.println(s.name);
    }
}
```

Terminal Output

```
$ java Program
Exception in thread "main" java.lang.NullPointerException: Cannot read field "name" because "s" is
null
    at exceptions.NPE.main(NPE.java:8)
```


9. StringIndexOutOfBoundsException

Code

```
package exceptions;
class StringIndexOutOfBounds_Demo {
    public static void main(String args[]) {
        try {
            String a = "My name is likitha";
            char c = a.charAt(20);
            System.out.println(c);
        }
        catch(StringIndexOutOfBoundsException e) {
            System.out.println("StringIndexOutOfBoundsException");
        }
    }
}
```

Terminal Output

```
$ java Program
StringIndexOutOfBoundsException
```

10. UnsupportedOperationException

Code

```
package exceptions;
import java.util.Arrays;
import java.util.List;
public class UnException {
    public static void main(String[] args) {
        List<Integer> list = Arrays.asList(10, 20, 30);
        try {
            list.add(40);
        }
        catch (UnsupportedOperationException e) {
            System.out.println("Add operation is not supported");
        }
    }
}
```

Terminal Output

```
$ java Program
Add operation is not supported
```